Example 12.24

Urn-I contains 8 red and 4 blue balls and urn-II contains 5 red and 10 blue balls. One urn is chosen at random and two balls are drawn from it. Find the probability that both balls are red.

Solution

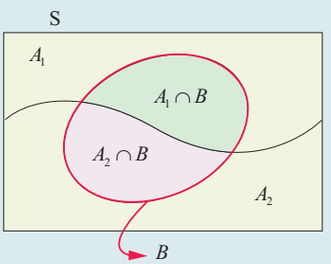
	Red balls	Blue balls	Total
Urn-I	8	4	12
Urn-II	5	10	15
Total	13	14	27

Let A_1 be the event of selecting urn-I and A_2 be the event of selecting urn-II.

Let B be the event of selecting 2 red balls.

We have to find the total probability of event B. That is, $P(B)$.

Clearly A_1 and A_2 are mutually exclusive and exhaustive events.



We have

$$P(A_1) = \frac{1}{2}, \quad P(B/A_1) = \frac{8c_2}{12c_2} = \frac{14}{33}$$

$$P(A_2) = \frac{1}{2}, \quad P(B/A_2) = \frac{5c_2}{15c_2} = \frac{2}{21}$$

We know

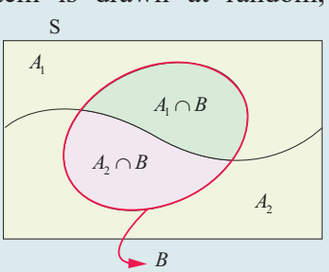
$$P(B) = P(A_1) \cdot P(B/A_1) + P(A_2) \cdot P(B/A_2)$$

$$P(B) = \frac{1}{2} \cdot \frac{14}{33} + \frac{1}{2} \cdot \frac{2}{21} = \frac{20}{77}.$$

Example 12.25

A factory has two machines I and II. Machine-I produces 40% of items of the output and Machine-II produces 60% of the items. Further 4% of items produced by Machine-I are defective and 5% produced by Machine-II are defective. If an item is drawn at random, find the probability that it is a defective item.

Solution



Let A_1 be the event that the items are produced by Machine-I, A_2 be the event that items are produced by Machine-II. Let B be the event of drawing a defective item.

We have to find the total probability of event B. That is, $P(B)$.

Clearly A_1 and A_2 are mutually exclusive and exhaustive events.

Therefore, $P(B) = P(A_1) \cdot P(B/A_1) + P(A_2) \cdot P(B/A_2)$

We have $P(A_1) = 0.40, \quad P(B/A_1) = 0.04$

$P(A_2) = 0.60, \quad P(B/A_2) = 0.05$

$$P(B) = P(A_1) \cdot P(B/A_1) + P(A_2) \cdot P(B/A_2)$$

$$= (0.40)(0.04) + (0.60)(0.05)$$

$$= 0.046.$$

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